Rate–Distortion Analysis of Event Streams from Moving Edges

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***Abstract*—An event camera reports when the log intensity of a pixel crosses a threshold. When an edge moves across the sensor, the pixels on its path share the edge and its motion, but each fires on its own lattice of threshold levels, offset by a threshold phase, and every event carries timing jitter. We derive the rate-distortion function of the event times under a mean-squared timing tolerance, given the edge and its motion, for a ramp edge that passes each pixel once under an affine flow with independent uniform phases and Gaussian jitter. Below the jitter variance a code describes every event time, and the rate is exact. Up to $D_A = \sigma^2 + \tau^2/12$, where τ is the event spacing of the pixel, it describes one number per pixel, and the rate is known within 0.255 bit. From D_A on the conditional rate is zero, and describing the motion of a path of M pixels adds $O(\ln M)$ nats. A sender that instead moves reference events along the motion keeps the phase of the source pixel: for equal spacings the timing error doubles, and on a voxel grid the moved counts reproduce a share of about Δ_t/τ of the true counts when the bin Δ_t is shorter than τ. A controlled model shows that the phases alone cause this loss. On recordings from a static and a moving camera, a per-block test detects no phase lock on regular event trains with spacings below 4 ms, and a lock appears only in a band consistent with mains lighting. Coding count pictures with HEVC shows the consequences: on fine time grids its P pictures save bits against intra pictures but not against a list of the counts, its lossy decodes remove single events, and the motion recovered from them falls with the rate.**

***Index Terms*—Event camera, send-on-delta sampling, level-crossing sampling, rate-distortion theory, generalized sampling, motion compensation, video coding, HEVC, neuromorphic sensing.**

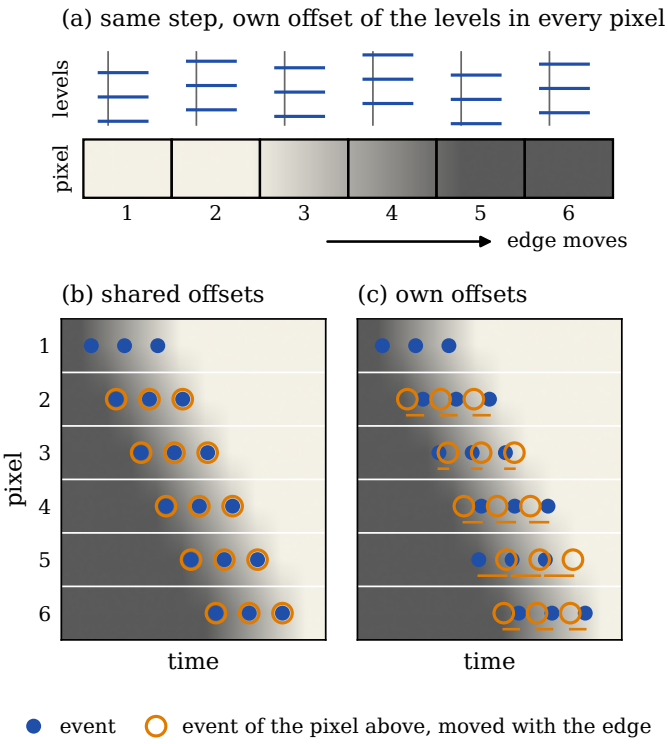


Fig. 1. An edge crosses a row of pixels. (a) Every pixel has levels of the same step and its own offset, the threshold phase. (b), (c) The events of each pixel in time, and those of the pixel above moved along the motion, for shared and for independent phases. Jitter is omitted.

I. Introduction

An event camera has no frames. Each pixel compares its log intensity with a stored reference and emits an event, with a time stamp and a polarity, whenever the difference reaches a contrast threshold [1], [2]. The output is a stream of address-events whose rate follows the scene: it is close to zero for a static scene and rises with the speed and the contrast of what moves. A sensor node that forwards this stream over a rate-limited link must decide how many bits the stream needs, and a receiver that does not need every event at its exact time should let it send fewer. How many bits does the stream of a moving scene need when the receiver tolerates a given timing error?

The stream of a moving scene looks highly redundant. If an object moves across the sensor, the pixels along the path of one of its edges observe the same intensity profile, one after the other. Yet they do not emit the same events. Each pixel compares the intensity with its own reference, so it fires on its own lattice of levels, and the offset of that lattice, the threshold phase of the pixel, differs from pixel to pixel. The same profile therefore produces a different event train at every pixel (Fig. 1). Moving the events of one pixel to the next along the motion of the edge removes the delay between the two arrivals, but not the offset between the two lattices (Fig. 1c). Which of these differences a code must describe depends on the timing error that the receiver tolerates.

Existing event coders approach the rate empirically. Lossless coders exploit the spatial and temporal correlation of addresses and time stamps [3]–[5]. Lossy coders discard or aggregate events [6], [7], and the coder of [8] suppresses events that an on-sensor optical flow can predict and lets the receiver extrapolate them. Event streams have also been aggregated into count pictures and coded by HEVC [4], [9], a video coder whose prediction is motion compensated [10]. Reported results are hard to compare [11], [12]. On the theoretical side, rate-distortion results for point processes concern Poisson and finite point processes [13]–[16]. Sampling theory comes closer to the mechanism. An event pixel is a send-on-delta sampler [17], which for a monotone input coincides with level-crossing sampling [18], [19], and an object that crosses M pixels is

sampled by M such samplers. This recalls the generalized sampling expansion [20] and multi-channel time encoding, where shifted integrate-and-fire encoders share one input and the shifts make the channels complementary [21]–[23]; the threshold phases play the role of these shifts. None of these results separates the cost of what the pixels of a path share, the edge and its motion, from the cost of what each pixel holds privately, and so none says what a timing tolerance allows a code to leave out.

We model the events of a ramp edge that passes each pixel once under an affine flow, with independent uniform phases and Gaussian jitter, and give the edge and its motion to the encoder and the decoder. The rate-distortion function of the event times under a mean-squared timing tolerance then has three regimes. Below the jitter variance a code describes every event time. Up to a threshold $D_A = \sigma^2 + \tau^2/12$, set by the event spacing τ of the pixel, it describes one number per pixel, its phase together with its mean jitter. Beyond D_A it describes nothing: the decoder regenerates the events from the edge and the motion. Describing the motion itself adds $O(\ln M)$ nats along a path of M pixels. A sender holds the events and not the edge, and a common substitute is to move reference events along the motion, as the prediction of a video coder does. A moved event keeps the phase of its source pixel, and the same event spacing that sets D_A sets the time bins over which moved counts remain predictive. A controlled model, a phase test on recordings and an HEVC coder test and illustrate these consequences.

1) The conditional rate-distortion function of the event times. Under independent phases and jitter, the event times of the pixels are conditionally independent given the edge and the motion, and the lossless code length of their time stamps grows in proportion to M (Theorem 1). The rate-distortion function of a pixel is exact below the jitter variance, known within 0.255 bit up to D_A, and zero from D_A on (Theorem 2); a path allocates the tolerance over its pixels (Theorem 3), and a motion that must be described costs $O(\ln M)$ nats (Proposition 1).

2) The cost of reusing events along the motion. For different source and target pixels with equal event spacings, moved events have twice the timing error of the regenerated ones (Proposition 2), and on a voxel grid the counts of a source pixel moved along the true motion reproduce a share of about Δ_t/τ of the true counts when the bin is shorter than the event spacing. A controlled model, with exact motion and equal counts, shows that the phases alone cause this loss.

3) Recordings and a video coder. On recordings from a static camera [24] and from a moving car [25], a per-block phase test detects no lock of moved events on the regular event trains where it can be applied, and a reconstruction shows what a receiver loses when the timing is left out. Coded by x265, an unmodified HEVC encoder, count pictures on fine time grids show a prediction gain against intra pictures but no gain against a list of the counts, lossy errors that remove single events, and a motion task that degrades with the rate.

Section II states the model and the problem. Section III gives the structure of the events, Section IV the rate-distortion function, and Section V the cost of moved events. Section VI reports the experiments. Proofs are in the appendices. Rates are in nats unless bits are named, $\|\cdot\|$ is the Euclidean norm of a vector and the spectral norm of a matrix, and $[x]^+ = \max(x, 0)$.

II. Model

We first state the pixel rule, from which the phase mechanism follows for any scene. We then specialize to a ramp edge in uniform passage, for which the rates have closed forms.

A. Pixel

Pixels sit at the lattice points $i \in \mathbb{Z}^2$ of the image plane. Pixel i observes the log intensity $\ell_i(t) = L(i,t)$ of a continuous field L.

(A1) Send-on-delta. Pixel i holds a reference μ_i with $|\ell_i - \mu_i| < C$ at the start. It emits an event at the first time t at which $|\ell_i(t) - \mu_i| = C$, with polarity $p = \operatorname{sign}(\ell_i(t) - \mu_i)$, and then sets $\mu_i \leftarrow \mu_i + pC$. The contrast threshold $C > 0$ is common to all pixels and known to the decoder. The *phase* of the pixel is the number $a_i \in (0, C]$ with $\mu_i \in a_i + C\mathbb{Z}$ at the start.

B. Scene and Motion

A template $g : \mathbb{R}^2 \to \mathbb{R}$ gives the log intensity in object coordinates. At time t the object point u is at the image position $\phi_t(u) = J_t u + b_t$, with $J_t \in \mathbb{R}^{2\times 2}$ invertible, so

$$L(x,t) = g\big(\phi_t^{-1}(x)\big) \tag{1}$$

and pixel i observes

$$\ell_i(t) = g\big(u_i(t)\big), \qquad u_i(t) = J_t^{-1}(i - b_t). \tag{2}$$

The curve u_i is the trajectory of pixel i in object coordinates. The velocity field $v(x,t) = \dot\phi_t(\phi_t^{-1}(x))$ is affine in x and satisfies the brightness constancy relation

$$\frac{\partial L}{\partial t}(x,t) + \langle \nabla L(x,t), v(x,t)\rangle = 0, \tag{3}$$

which follows from differentiating $L(\phi_t(u), t) = g(u)$ in t. We call the motion an *affine flow* when the velocity field is also constant in time,

$$v(x) = \Lambda x + v_0, \tag{4}$$

with generator $\Lambda \in \mathbb{R}^{2\times 2}$. A translation has $\Lambda = 0$. The trace of Λ is the rate at which the object grows or shrinks in area, its antisymmetric part rotates the object, and its symmetric traceless part shears it. At a given instant, any smooth velocity field has the form (4) to first order in x around a point, so (4) is the local model of a general motion over a short time.

C. Edge

(A2) Ramp edge. The template is a ramp across a line with unit normal ν,

$$g(u) = \varrho\big(\langle \nu, u\rangle - c_0\big), \qquad \varrho(z) = \begin{cases} 0, & z \le 0, \\ \gamma z, & 0 \le z \le w, \\ V, & z \ge w, \end{cases} \tag{5}$$

with slope $\gamma > 0$, width w and height $V = \gamma w = nC$ for an integer $n \geq 1$.

Pixel i sees the edge through its edge coordinate

$$z_i(t) = \langle \nu, u_i(t) \rangle - c_0, \qquad \ell_i(t) = \varrho\big(z_i(t)\big). \tag{6}$$

(A2′) Uniform single passage. The edge passes pixel i once: $z_i < 0$ before a time t_i, $z_i > w$ after the passage, and in between z_i is affine in t with slope $r_i > 0$.

Inside the ramp, $\nabla L = \gamma \nu_t$ with $\nu_t = J_t^{-\top} \nu$ a normal of the edge in the image, and $\dot{z}_i = -\langle \nu_t, v(i,t) \rangle$. By (3), the intensity of pixel i rises during the passage at the rate

$$\gamma r_i = -\langle \nabla L, v \rangle = \|\nabla L\| \, v_{\perp,i}, \tag{7}$$

with the field evaluated at pixel i. Here $v_{\perp,i} = -\langle \nabla L, v \rangle / \|\nabla L\|$ is the speed at which the edge advances over the pixel along its normal, counted positive when the intensity of the pixel rises, as it does in a passage with $r_i > 0$. We call

$$\tau_i = \frac{C}{\gamma r_i} = \frac{C}{\|\nabla L\| \, v_{\perp,i}} \tag{8}$$

the *event spacing* of pixel i. It is the time that the ramp takes to pass the pixel, divided by n. At a fixed image slope $\|\nabla L\|$ it is inversely proportional to the normal speed of the edge. Assumption (A2′) holds exactly for a translation, where τ_i is the same for all pixels, and for a plane that approaches the camera along the optical axis at constant speed, where J_t^{-1} is affine in t and b_t is fixed. For an affine flow it holds to first order (Remark 3), and τ_i varies from pixel to pixel.

D. Phase, Jitter and Clock

(A3) Phase. Before the edge arrives, pixel i rests at the level 0 with its reference behind the passage, $\mu_i \in (-C, 0]$, so that $a_i = \mu_i + C$. The phases a_i are independent across pixels, uniform on $(0, C]$, and independent of the jitter.

(A4) Jitter and clock. The kth event of pixel i is recorded at $T_{i,k} = t_{i,k} + \varepsilon_{i,k}$, where $t_{i,k}$ is the time given by (A1) and the $\varepsilon_{i,k}$ are independent $\mathcal{N}(0, \sigma^2)$ variables, independent of the phases, with $\sigma > 0$. The stored value is $Y_{i,k} = \delta \lfloor T_{i,k}/\delta \rfloor$ for a clock resolution δ. The index k of each recorded event is known.

The uniform phase does not arise inside the model, in which every pixel rests at the same level. It stands for the offsets of real pixels and for their earlier activity. A pixel that the edge crosses from the high level to the low level is covered by the mirrored assumption, with its reference in $[V, V + C)$. Remark 2 treats a reference that is ahead of the passage.

We write $T_i = (T_{i,1}, \ldots, T_{i,n})$ and $Y_i = (Y_{i,1}, \ldots, Y_{i,n})$, and $\mathcal{P}$ for the set of pixels that the edge passes during the observation, with $M = |\mathcal{P}|$. Let $\eta(a, \sigma)$ denote the differential entropy of the sum of a uniform variable on $(0, a)$ and an independent $\mathcal{N}(0, \sigma^2)$ variable. It satisfies

$$\tfrac{1}{2} \ln\big(a^2 + 2\pi e \sigma^2\big) \leq \eta(a, \sigma) \leq \tfrac{1}{2} \ln\big(2\pi e (a^2/12 + \sigma^2)\big) \tag{9}$$

by the entropy power inequality [26, Ch. 17] and by the Gaussian bound on the entropy at a given variance [26, Ch. 8], and $\eta(a, \sigma) - \ln a \to 0$ as $\sigma / a \to 0$.

E. Problem

The encoder and the decoder know the template g and the flow ϕ, and hence the pixels that the edge passes, the number n of events of each and their polarity. The index k of every event is known (A4). The addresses and the correspondence between events are therefore given, and what remains to be described are the event times. Theorem 1 counts the bits of their time stamps at the clock resolution δ. Theorems 2 and 3 count the bits of the event times under a mean-squared timing tolerance, and their zero rate is a rate given all of the above. Proposition 1 extends them to a flow that must be described, while the decoder still holds the template. A practical sender holds neither, and Section V turns to what it can send instead.

III. Structure of the Events

Lemma 1. *Under (A1), for any continuous ℓ_i, every event of pixel i occurs at a time t with $\ell_i(t) \in a_i + C\mathbb{Z}$.*

The lattice $a_i + C\mathbb{Z}$ never changes. The phase is a property of the pixel, set by its history and not by the edge that passes.

Lemma 2. *Under (A1), (A2), (A2′) and $\mu_i \in (-C, 0]$ before the passage, pixel i emits exactly n events, all of positive polarity, at*

$$t_{i,k} = t_i + A_i + (k-1)\tau_i, \qquad k = 1, \ldots, n, \tag{10}$$

where $A_i = a_i/(\gamma r_i)$. Under (A3), A_i is uniform on $(0, \tau_i]$.

Given the template and the flow, the noise-free events of a pixel are thus a function of one number, its phase (Fig. 1a). In the object coordinate, an event is a point of the graph of the profile on the level lattice of its pixel: the M pixels of a path sample the same profile on M lattices with independent offsets.

Theorem 1 (Lossless code length)**.** *Under (A1) to (A4) and (A2′), the vectors T_i, $i \in \mathcal{P}$, are independent given the template and the flow,*

$$h(T_i) = \frac{n-1}{2} \ln(2\pi e \sigma^2) + \eta(\sqrt{n}\,\tau_i, \sigma), \tag{11}$$

and

$$H(Y_i \mid g, \phi) = n \ln \frac{1}{\delta} + h(T_i) + e_\delta, \tag{12}$$

where $e_\delta \geq 0$ for every δ, and $e_\delta \to 0$ as $\delta/\sigma \to 0$, uniformly in τ_i.

When $\sigma \ll \tau_i$ and $\delta \ll \sigma$, (12) gives the cost per event

$$\frac{H(Y_i \mid g, \phi)}{n} \approx \frac{n-1}{n} \ln \frac{\sigma\sqrt{2\pi e}}{\delta} + \frac{1}{n} \ln \frac{\sqrt{n}\,\tau_i}{\delta}. \tag{13}$$

One event of each pixel carries the cost of the phase, and the other $n-1$ carry that of the jitter. The code length of the path is the sum of (12) over $\mathcal{P}$. It grows in proportion to M, and neither term of (13) falls as the path gets longer: a decoder that knows the edge and its motion exactly still needs (13) for every event. What the pixels of a path share, the edge and its motion, a code can describe once. What is private to each pixel, its phase and its jitter, it must describe for each pixel, and (12) is that irreducible conditional rate.

Remark 1 (Several edges). Theorem 1 counts one edge that passes each pixel once. If further edges of the same object pass the same pixel, they meet the same lattice, so the phase is paid once per pixel and not once per edge. With many edges per pixel the cost per event tends to the jitter term $\ln(\sigma\sqrt{2\pi e}/\delta)$, which does not depend on the path either.

IV. Rate-Distortion Function

Theorem 1 gives the entropy of time stamps on a clock of resolution δ, and the results of this section give the rate of a lossy reconstruction of the continuous event times. Both describe the same structure, the shared edge and motion and the private phase and jitter, but they are different quantities. The decoder outputs $\hat{T}_i \in \mathbb{R}^n$ for each pixel, and the distortion is the mean-squared timing error

$$d(T_i, \hat{T}_i) = \frac{1}{n}\|T_i - \hat{T}_i\|^2. \tag{14}$$

A. One Pixel

Let $R_i(D) = \inf I(T_i;\hat{T}_i)$, with the infimum over the conditional laws of $\hat{T}_i$ that satisfy $\mathbb{E}\, d(T_i,\hat{T}_i) \le D$, be the rate-distortion function of the events of pixel i. It is the rate that block codes over many independent passages approach. Define

$$s_i^2 = \frac{n\tau_i^2}{12} + \sigma^2, \qquad D_{A,i} = \sigma^2 + \frac{\tau_i^2}{12}, \tag{15}$$

$$\Gamma_i = \tfrac{1}{2}\ln(2\pi e\, s_i^2) - \eta(\sqrt{n}\,\tau_i, \sigma). \tag{16}$$

By (9), $0 \le \Gamma_i \le \frac{1}{2}\ln(2\pi e/12)$, which is 0.255 bit, for every σ and τ_i.

Theorem 2 (Rate-distortion function of a pixel)**.** *Under (A1) to (A4) and (A2′),*

$$R_i(D) = h(T_i) - \frac{n}{2}\ln(2\pi e D) \qquad \textit{for } 0 < D \le \sigma^2, \tag{17}$$

$$\big[U_i(D) - \Gamma_i\big]^+ \le R_i(D) \le U_i(D) \qquad \textit{for } \sigma^2 < D < D_{A,i}, \tag{18}$$

$$R_i(D) = 0 \qquad \textit{for } D \ge D_{A,i}, \tag{19}$$

with

$$U_i(D) = \frac{1}{2}\ln\frac{s_i^2}{nD - (n-1)\sigma^2}. \tag{20}$$

Moreover,

$$R_i(D) \ge \frac{n\,(D_{A,i} - D)}{2s_i^2} \qquad \textit{for } 0 < D \le D_{A,i}, \tag{21}$$

so that $R_i(D) > 0$ *for* $D < D_{A,i}$.

The three regimes have a direct reading. Below the jitter variance, every bit of timing precision costs a bit, per event: the rate falls by $\log_2 10 = 3.32$ bits per event and per decade of the root-mean-square tolerance $\sqrt{D}$. Between σ^2 and $D_{A,i}$, the $n-1$ jitter components orthogonal to a common shift of the events are no longer described, and the code describes one number per pixel, the phase together with the mean jitter, to within 0.255 bit. From $D_{A,i}$ on, nothing is described. The decoder that attains (19) outputs the mean times

$$\hat{T}_{i,k} = t_i + \big(k - \tfrac{1}{2}\big)\tau_i, \qquad k = 1, \dots, n. \tag{22}$$

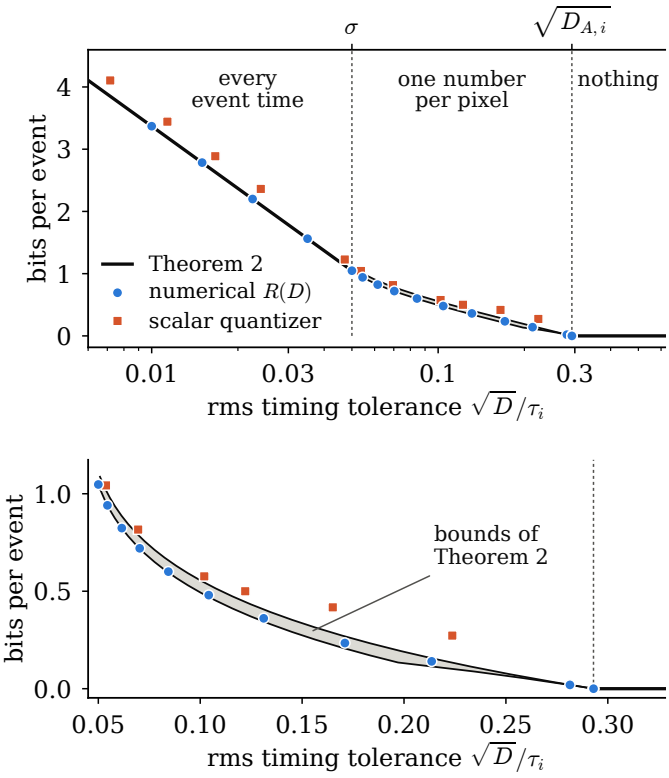


Fig. 2. Rate of one pixel against the timing tolerance for $n = 3$ and $\sigma/\tau_i = 0.05$: Theorem 2, a numerical rate-distortion function and a scalar quantizer. Left of σ every event time is described, between σ and $\sqrt{D_{A,i}}$ one number per pixel, and beyond $\sqrt{D_{A,i}}$ nothing. Bottom: the middle regime.

It *regenerates* the events of the pixel from the edge and the flow. Each regenerated event is within $\tau_i/2$ of its noise-free time, and the count and the polarities are exact.

Two of these statements extend beyond the ramp. If the noise-free event times of a pixel are any function of its phase, (17) holds below the jitter variance with the entropy of the actual event times, and the smallest distortion at zero rate is $\frac{1}{n}\operatorname{tr}\operatorname{Cov}(T_i)$, attained by the mean times. The middle regime, in which one number per pixel is described to within 0.255 bit, and the closed forms (11), (20) and (15) belong to the ramp in uniform passage, where the phase shifts all events of a pixel by a common time.

The threshold $D_{A,i}$ grows with the square of the event spacing. By (8), a pixel whose intensity changes slowly has a large spacing and needs a large tolerance before its events can be regenerated, and a pixel that is crossed quickly needs a small one.

B. Numerical Check

Fig. 2 compares Theorem 2 with a rate-distortion function that is computed without it, for one pixel with $n = 3$ and $\sigma/\tau_i = 0.05$: the iteration of [27] on the phase component X_i of (34), with the $n-1$ normal components in closed form at the same slope.

Below σ^2 the numerical rate equals (17) to within 4×10^{-6} nat. Between σ^2 and $D_{A,i}$ it stays inside the band between the bounds (18) and (21), which is at most $\Gamma_i = 0.187$ bit per pixel

wide here, against the limit of 0.255 bit. Near σ^2 the phase is described finely and the rate sits on the lower bound, which uses its uniform distribution; toward $D_{A,i}$ the description is coarse and the rate meets (21), which uses only its variance. From $D_{A,i}$ on the rate is zero, and the regenerated events (22) of 2×10^6 simulated pixels have the distortion $0.9994\, D_{A,i}$.

The squares are a uniform scalar quantizer with entropy coding, applied to what each regime describes. On every component of the event times it stays 0.254 to 0.258 bit per event above (17), the loss $\frac{1}{2}\ln(2\pi e/12)$ of a uniform quantizer at fine resolution. On the mean event time alone, with the other components left out, it stays 0.09 to 0.19 bit per event above the lower bound up to $D_{A,i}$. A code that describes every event time, then one number per pixel, then nothing, is therefore within 0.26 bit per event of the limit in all three regimes.

C. A Path

For the M pixels of a path, let the distortion be the average of (14) over $i \in \mathcal{P}$, and let $R_{\mathcal{P}}(D)$ be the rate-distortion function of $(T_i)_{i\in\mathcal{P}}$ under it.

Theorem 3 (Rate-distortion function of a path)**.** *Under (A1) to (A4) and (A2′),*

$$R_{\mathcal{P}}(D) = \min\Big\{ \sum_{i\in\mathcal{P}} R_i(D_i) : \ \frac{1}{M}\sum_{i\in\mathcal{P}} D_i \le D \Big\}. \tag{23}$$

In particular,

$$R_{\mathcal{P}}(D) = \sum_{i\in\mathcal{P}} h(T_i) - \frac{Mn}{2}\ln(2\pi e D) \qquad \text{for } 0 < D \le \sigma^2, \tag{24}$$

and $R_{\mathcal{P}}(D) = 0$ if and only if

$$D \ \ge \ \bar{D}_A = \sigma^2 + \frac{1}{12M}\sum_{i\in\mathcal{P}} \tau_i^2. \tag{25}$$

If $\tau_i \le \tau_{\max}$ on the path, then for $D \le \bar{D}_A$

$$R_{\mathcal{P}}(D) \ge \frac{nM\,(\bar{D}_A - D)}{2\,(n\tau_{\max}^2/12 + \sigma^2)}. \tag{26}$$

If the tolerance is imposed on every pixel separately, the rate is zero if and only if $D \ge \max_i D_{A,i}$, and the slowest crossing sets the tolerance. By (26), below $\bar{D}_A$ a rate that is proportional to M remains.

D. The Motion

Theorems 2 and 3 assume that the decoder knows the flow. We now extend them to a flow that must be described, while the decoder still holds the template. An error in a speed displaces the regenerated events in proportion to the distance that the edge has traveled, so a longer path needs the motion to a finer precision for the same distortion. A parameter of fixed dimension costs only logarithmically more at a finer precision, and Proposition 1 states this in general form.

Proposition 1 (Description of the motion)**.** *Let the flow depend on a parameter $\theta = (\theta_1, \dots, \theta_q)$ that ranges over a box whose jth side has length at most $L_j M^{\alpha_j}$, and let the mean times $\bar{t}_{i,k}(\theta) = t_i(\theta) + (k - \frac{1}{2})\tau_i(\theta)$ of (22) satisfy $|\partial \bar{t}_{i,k}/\partial\theta_j| \le S_j M^{\beta_j}$ on the box, for all events of the path, with constants $L_j, S_j > 0$ and exponents $\alpha_j, \beta_j \ge 0$ that do not depend on M. Then for every $\xi > 0$ the events of the path can be described with average distortion at most $(1+\xi)\bar{D}_A$ using*

$$B \le \Big(\sum_{j=1}^{q} (\alpha_j + \beta_j)\Big)\ln M + O(1) \quad \textit{nats}, \tag{27}$$

where the $O(1)$ term depends on q, ξ, σ and the constants and not on M.

For a translation along a line with a speed in a fixed interval that excludes zero, θ holds the position of the edge, with a range of order M and a sensitivity of order one, and the speed, with a range of order one and a sensitivity of order M, so $B \le 2\ln M + O(1)$. The same holds with finite exponents, and hence $B = O(\ln M)$, for an affine flow whose normal speed is bounded away from zero on the path and whose duration is polynomial in M. The lossless cost of the same events is the sum of (12), of order M, so the ratio of the two grows at least as $M/\ln M$. Above $\bar{D}_A$ the description that remains is $O(\ln M)$. Below it, a rate proportional to M remains. The bound (27) is for a receiver that holds the edge.

Remark 2 (Departures from the model)*. Reference ahead of the passage.* If $\mu_i \in (0, C)$ before a rising passage, as after a passage of the opposite polarity at the same pixel, the first level that the intensity meets is the reference itself, which does not fire. The pixel then emits $n-1$ events, and the results hold with $n-1$ in place of n and $t_i + \tau_i$ in place of t_i. *Height.* If the height of the edge is not a multiple of the threshold, $V/C = n + f$ with $0 < f < 1$, pixel i emits $n+1$ events when $a_i \le fC$ and n otherwise. Reproducing the count exactly costs the binary entropy of f per pixel, and a zero-rate statement then holds only for a distortion measure that tolerates one unmatched event per pixel, with a modified threshold. *Sensor.* Noise events, threshold mismatch between pixels, a refractory time and a jitter that depends on the slope are outside the model, and their effect on the three regimes is not analyzed here.

V. Reusing Events Along the Motion

The decoder (22) needs the edge and the flow. A sender has neither. It has the events. It could estimate the edge from them, and this paper does not pursue that route. We study a common substitute: the receiver moves reference events that it already holds along the motion. This is motion-compensated prediction, the core of a hybrid video coder [9], [10]: a block of a decoded picture, displaced by a motion vector, predicts a block of the next picture, and the coder sends the residual. This section gives what a moved event costs in timing error and in the fidelity of a receiver that counts events.

A. Timing Error

Proposition 2 (Moved events)**.** *Let the same edge pass pixel j in an earlier window and a different pixel i later, under (A1) to (A4) and (A2′), and let the motion be exact, so that*

the kth event of pixel j is moved to pixel i at the time $\tilde{T}_{i,k} = T_{j,k} + t_i - t_j$. Then

$$\frac{1}{n}\,\mathbb{E}\|T_i - \tilde{T}_i\|^2 = D_{A,i} + D_{A,j} + \frac{4n^2-1}{12}\,(\tau_i - \tau_j)^2. \quad (28)$$

For different source and target pixels with equal event spacings the error is $2D_A$, twice that of (22): a moved event carries the phase and the jitter of its source pixel, and both are independent of those of its target. Unequal spacings, as under a change of scale, add the last term, which grows with the square of the number of events per pixel. If the moved events fall on their own pixel, as for an object at rest, the phases coincide and only the jitter term $2\sigma^2$ remains.

B. Voxel Fidelity

The timing error (14) presupposes that predicted and true events are paired. Receivers of event streams count events instead, on a voxel grid: the number of events of each polarity in blocks of $\Delta_x \times \Delta_x$ pixels and in time bins of length Δ_t. A video coder codes the same counts as pictures. For a reconstruction with voxel counts c' and a true set with counts c, of sizes N' and N, the fidelity is

$$F = \frac{2\sum \min(c', c)}{N' + N} = 1 - \frac{\sum |c' - c|}{N' + N}, \quad (29)$$

the share of the events of the two sets that can be paired one to one inside a voxel.

For blocks of one pixel, a train that is long against the event spacing, and a jitter that is small against the bin, Lemma 2 gives the fidelity of the regenerated events (22) directly: a bin holds at most one event of each train when $\Delta_t \le \tau_i$, the offset between the two trains is uniform, and in expectation over the phase

$$F = \frac{\Delta_t}{\tau_i} \qquad \text{for } \Delta_t \le \tau_i, \quad (30)$$

with $F = 1$ at $\Delta_t = \tau_i$. The voxel counts of a pixel are thus reproduced only when the bin is at least as long as its event spacing, the same scaling as the threshold of Theorem 2, for which $\sqrt{D_{A,i}} = \tau_i/\sqrt{12}$ when $\sigma \ll \tau_i$.

Moved events do almost as well on this grid. With equal spacings and no jitter, the kth moved event lies at the offset $\delta = A_i - A_j$ from the kth true event, and δ has the triangular density $(\tau_i - |\delta|)/\tau_i^2$ on $(-\tau_i, \tau_i)$. Two events at the offset x share a bin with probability $[1 - |x|/\Delta_t]^+$ over the position of the bins. When $|\delta| > \tau_i - \Delta_t$, a moved event can share its bin with a neighbor of its true event instead, which has a partner for $n-1$ of the n events of a train. For a train of n events and $\Delta_t \le \tau_i/2$, the two contributions give

$$F = \frac{\Delta_t}{\tau_i} - \frac{\Delta_t^2}{3n\tau_i^2}. \quad (31)$$

A voxel grid does not distinguish the events of a pixel, so the doubled timing error of (28) lowers the fidelity only by a term of order $1/n$.

Equation (31) is the expected fidelity of the events of a given source pixel moved along the true motion. When the bin is shorter than the event spacing, the count of a pixel is zero or one, which bins hold the ones is decided by the phases, and about a share $1 - \Delta_t/\tau_i$ of the counts remains a residual. When the bin is long against the spacing, a bin holds whole trains or long parts of them, the offset δ moves only the events at their ends, and the moved counts approach the true ones. Prediction along the true motion therefore removes most of the counts on long bins and few on short ones. Equation (31) describes one way of reusing events, not a bound for all motion compensation. It makes three predictions that Section VI tests: with the motion exact, the phases alone make moved counts miss; on recordings, the positions of moved events within the trains of their target pixels show no lock; and a coder that predicts counts along the motion gains little where Δ_t is short against τ.

VI. Experiments

The experiments test the predictions of Section V and then follow them into a standard coder. On the model, a controlled experiment isolates the phase as the cause of the missing counts (Section VI-B). On recordings, a phase test checks the premise of the mechanism on the events where it can be measured (Section VI-C), and a reconstruction shows what a receiver loses when a code leaves out timing detail (Section VI-D). An HEVC coder then serves as a case for what the mechanism means in practice (Section VI-E).

A. Protocol

Data. The recordings from a static camera are the 112 recordings of the training split of eTraM [24], 61 by day and 51 by night, from a sensor of 1280×720 pixels above a road, with 2.4×10^{10} events. The recordings from a moving camera are the 41 training sequences of DSEC [25], left event camera, 640×480 pixels, on a car, with 3.7×10^{10} events in 2634 s. Twelve eTraM recordings and five DSEC sequences are held out for tuning, and the results below are over the other 100 and 36.

Grids. Time is cut into windows of $T_0 = 33.3$ ms and each window into k slices, so $\Delta_t = T_0/k$. A voxel holds one pixel and one polarity. We use $k \in \{1, 3, 6, 12, 32, 64\}$, from $\Delta_t = 33.3$ ms to 0.52 ms. A group is four consecutive windows, taken every 3 s of an eTraM recording and every 2 s of a DSEC sequence.

Video coder. The counts of a group form one clip of $4k$ pictures, with the two polarities side by side in the luma plane and constant chroma. They are coded by x265 3.5 [28], an encoder of HEVC [9], through ffmpeg 4.4.2, in the Main profile at the preset `slow`, with one intra picture and P pictures after it, without B pictures, and with adaptive quantization and the psychovisual options off. In lossless mode the luma is the count. At a constant quantization parameter (QP) the luma is 16 times the count, the decoder rounds the luma divided by 16, and the intra picture is coded at the QP of the P pictures. Counts are clipped at 15, which removes less than 1% of the events for $k \ge 3$; clipped events count as lost, and the Supplementary Material gives the clip on other grids. Every bit of the stream is counted, parameter sets included. A lossless decode returns the clipped counts, not the event stream.

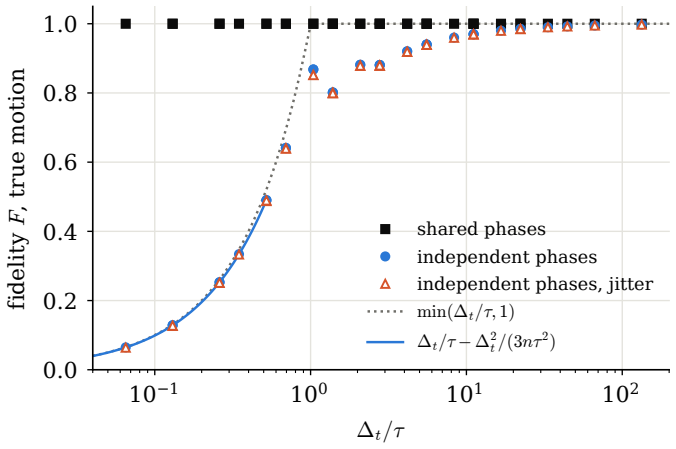


Fig. 3. Fidelity of the counts moved along the exact motion on the model, $n = 3$, one-pixel voxels, for τ from 0.25 to 8 ms and every k.

Voxel list. As a coder without prediction, the voxel list sorts the events of a window by voxel, writes the gaps between their voxel indices in a variable-length integer code and compresses the result with Zstandard at level 19 [29]. It codes the same clipped counts and returns them exactly.

Motion task. In each group, the ten densest squares of 40×40 pixels with at least 200 events in the third window are selected, and the velocity of each is estimated by contrast maximization [30] on the true events and on the decoded events. A square counts as recovered when the two velocities differ by at most one pixel per window, and a square that the decode leaves without events counts as not recovered.

Statistics. Bits are per true event. A recording enters with the sums over its groups. We report the median over recordings, or the value pooled over recordings with a 95% interval from a bootstrap over recordings where one is given. The coder uses one group in 40 of eTraM and one in 20 of DSEC (42 recordings by day, 34 by night and 36 sequences), and the phase test one group in five (55, 45 and 36).

B. The Phase as the Cause: a Controlled Model

Straight edges cross a sensor of 320×240 pixels at six pixels per window, 40 pixels apart, for twelve windows, with $n = 3$ and τ from 0.25 to 8 ms. Three conditions share the edges, their arrival times and the number of events of every pixel: phases shared along each path without jitter, independent phases without jitter, and independent phases with $\sigma = 0.05\,\tau$. The prediction of a window is the counts of the previous window moved by exactly six pixels, so the motion is exact and only the phases and the jitter can make the counts differ.

With shared phases the moved counts equal the true ones, $F = 1$ in all 36 cells (Fig. 3). With independent phases they follow (31) to within 4×10^{-4} wherever it applies, and F falls with Δ_t/τ: at $\tau = 1$ ms, from 0.99 at $\Delta_t/\tau = 33$ to 0.49 at 0.52. Jitter changes F by at most 0.016. Between $\Delta_t/\tau = 1$ and 4, where (30) would give 1 for regenerated events, the moved counts keep 0.80 to 0.92, since a moved train still sits at the offset of its source pixel; from $\Delta_t/\tau = 11$ on they keep 0.97 or more. In this model the phase alone causes the loss.

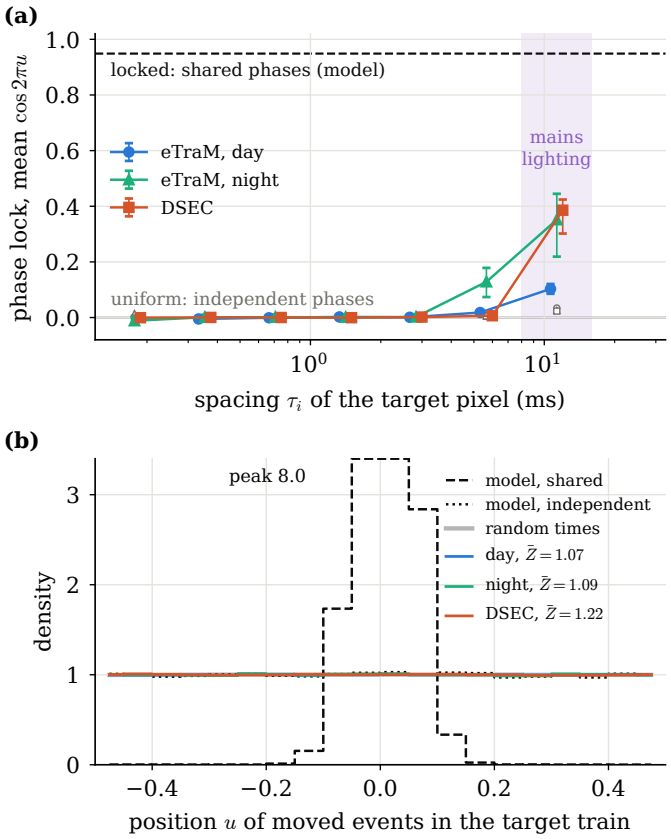


Fig. 4. Moved events on recordings, moving blocks. (a) Mean of $\cos 2\pi u_i$ against the spacing of the target pixel, with 95% intervals; open grey markers: random times. (b) Histogram of u_i for $0.25 \le \tau_i < 4$ ms, with the mean Rayleigh statistic per block; the peak of the shared-phase model is clipped.

C. Phases on Recordings

On recordings neither the motion nor the event spacing is known, so we test the premise of the mechanism rather than (31). For each block of 20×20 pixels, the shift that pairs the most events of window $w-1$ with window w is chosen, and the same shift, with the median offset between the first moved and the first true event of the pixels of the block, predicts window $w + 1$ from window w. A target pixel and polarity is tested when its true events in window $w + 1$ hold two consecutive intervals within a factor 1.2 of each other. With τ_i the median of such intervals, the position of the moved events in its train is

$$u_i = \frac{1}{2\pi} \arg\left(\sum_k e^{2\pi \mathrm{i}\, p_k/\tau_i} \sum_l e^{-2\pi \mathrm{i}\, t_l/\tau_i} \right), \qquad (32)$$

over its predicted times p_k and its true times t_l. Under (A3), $A_j - A_i$ is uniform modulo τ_i, so u_i is uniform on $[0, 1)$ whatever the residual offset of the motion and the length of the trains. If the pixels of a path shared their phases, u_i would be constant. Pooling $\cos 2\pi u_i$ over pixels detects a lock common to all blocks, but locks at different values in different blocks would cancel. The Rayleigh statistic of a block of m tested pixels,

$$Z = \frac{1}{m} \left| \sum_i e^{2\pi \mathrm{i}\, u_i} \right|^2, \qquad (33)$$

does not depend on the value of the lock: its mean is 1 for independent uniform u_i and close to m when the pixels of a block share one. Before the recordings were read, a 95% interval of the mean Z below 2 was set to read as no lock within blocks, one above 5 as a lock, and anything else as partial. Blocks with a zero shift are left out, since a pixel moved onto itself keeps its own phase.

For pixels with τ_i between 0.25 and 4 ms, the mean of $\cos 2\pi u_i$ lies within 0.006 of 0 in every bin of τ_i and in all three strata, against 0.91 to 0.97 for shared phases on the model (Fig. 4a). The histogram of u_i is flat (Fig. 4b). Per block, the mean Z is 1.07 (95% interval 1.02 to 1.13) by day, 1.09 (1.06 to 1.14) at night and 1.22 (1.04 to 1.46) on DSEC, over 15,028, 6,933 and 44,715 blocks; the same test gives 63.8 on the model with shared phases and 1.05 with independent ones. All three strata therefore read as no lock within blocks. A small excess over random times remains: their mean Z is 1.00, and 1.1% to 1.7% of the blocks exceed $Z = \ln 100$, against 0.8% to 1.0% for random times. The tested pixels hold 33%, 27% and 27% of the true events of the scored windows, and those in the band of Fig. 4b hold 29%, 20% and 19%. The other events come from pixels without a regular train in the window.

A lock appears at longer spacings: for τ_i between 8 and 16 ms the mean of $\cos 2\pi u_i$ reaches 0.10 by day, 0.35 at night and 0.39 on DSEC, and at night already 0.13 between 4 and 8 ms. That band holds the periods of mains lighting, 8.33 and 10 ms, and the lock is consistent with lamps that modulate many pixels in step, a common cause outside the model. On the selected regular trains and at the tested spacings, the recordings thus show no detectable lock of the phases along the motion.

On the same pixels, the counts moved along the estimated motion reproduce 0.1 to 0.4 of the true counts on the grid of $T_0/32$, above the 0.06 to 0.20 of random times. Count matching alone, without any timing, would reach 0.48 to 0.75 there, and the motion is estimated, so these values mix the phase with count mismatch and motion error and do not test (31).

D. What a Receiver Loses

Fig. 5 shows what a receiver displays as a code leaves out more of the timing, for a car that crosses the sensor at 13 pixels per window by day; a second car, at night at 9 pixels per window, gives the same picture. The three codes follow the order of Theorem 2 without attaining its rates. The first sends every event with its time in bins of $T_0/256$ by the voxel list. The second moves the events of the previous window along the motion of their block and corrects them with one number per pixel: the offset of the first true event of the pixel from its first moved event, rounded to $T_0/64$, or a symbol that deletes them. The third sends only the motion. The motion of each block is fitted on the window it predicts, so that the lower rows show what motion cannot carry and not the errors of its estimate.

Over the whole window the three reconstructions are hard to tell apart from the recording: the motion places the edges of the car where they are. In a slice of 2.1 ms the moved events fall on other pixels than the recorded ones, since each pixel fires on its own phase. Of the 4821 pixels with an event in the slice, all events reproduce 4816; the per-pixel correction reproduces 3098 and adds 1864 elsewhere; the motion alone reproduces 2569 and adds 3981. All events cost 9.8 bits per event by day and 9.5 at night and keep $F = 1$ on the grid of $T_0/32$. The per-pixel correction costs 2.4 and 2.7 bits per event and keeps $F = 0.48$ and 0.47; the motion alone costs 0.04 and 0.07 bits and keeps 0.28 and 0.25. On the grid of $T_0/3$ the three codes keep 1.00, 0.73 to 0.75 and 0.61 to 0.64: the longer the bin against the event spacing, the more of the recording the motion carries.

TABLE I
LOSSLESS BITS PER EVENT ON RECORDINGS, ONE-PIXEL VOXELS: X265 WITH P PICTURES (P), WITH INTRA PICTURES ONLY (I), AND THE VOXEL LIST (Q).

	eTraM, day			eTraM, night			DSEC		
k	P	I	Q	P	I	Q	P	I	Q
1	2.13	2.46	2.18	1.97	2.19	2.01	1.64	1.70	1.69
3	3.32	4.81	3.40	3.14	3.90	3.25	2.76	3.00	2.83
12	6.43	11.3	6.12	5.51	8.17	5.59	4.88	5.12	4.74
64	11.5	38.2	10.2	9.75	24.6	9.06	8.09	8.39	7.48

E. A Video Coder as a Case

HEVC on count pictures has been used to code event streams without loss [4]. We use x265 as a case: it has motion-compensated prediction, a transform and a quantizer, and we can compare its P pictures with intra pictures, which isolates the gain of prediction, and with the voxel list, which gives the total cost of a coder without prediction.

On the model. x265 codes the counts of Section VI-B, with independent phases, without loss (Fig. 6a). At $\tau = 1$ ms its P pictures cost 0.14 bit per event at $\Delta_t/\tau = 33$ and 7.9 at 0.52. They undercut the voxel list for $\Delta_t/\tau \geq 5.6$, match it near 2.8, and cost up to 1.7 times as much below 1.4. Against intra pictures only they save a factor of 2 to 2.5 at $\Delta_t/\tau \geq 11$ and of 1.1 to 1.3 below 1. The gain of temporal prediction shrinks where (31) says that moved counts stop predicting the true ones.

Without loss, on recordings. The P pictures need about the bits of the voxel list, and more on fine grids (Table I): the ratio of the two is 0.97 to 0.98 for $k \leq 3$, 0.99 to 1.05 at $k = 12$, and 1.08 to 1.13 at $k = 64$. Prediction itself still pays against intra pictures. On eTraM the intra pictures cost 1.1 to 3.4 times the P pictures, increasingly so at fine slices: a static camera leaves pictures almost empty, and the P pictures save the cost of signaling empty regions, which an intra picture pays anew. On DSEC, where the camera moves, the intra pictures cost 1.03 to 1.10 times the P pictures. The prediction gain thus survives on fine grids, but it does not make the coder cheaper than a list of the counts, as Theorem 1 and (31) lead one to expect: on fine grids the counts are set by the phases, which the motion does not carry.

With loss. On a fine grid almost all nonzero counts are one: 92% to 99% of the events lie in voxels of one event at $T_0/64$.

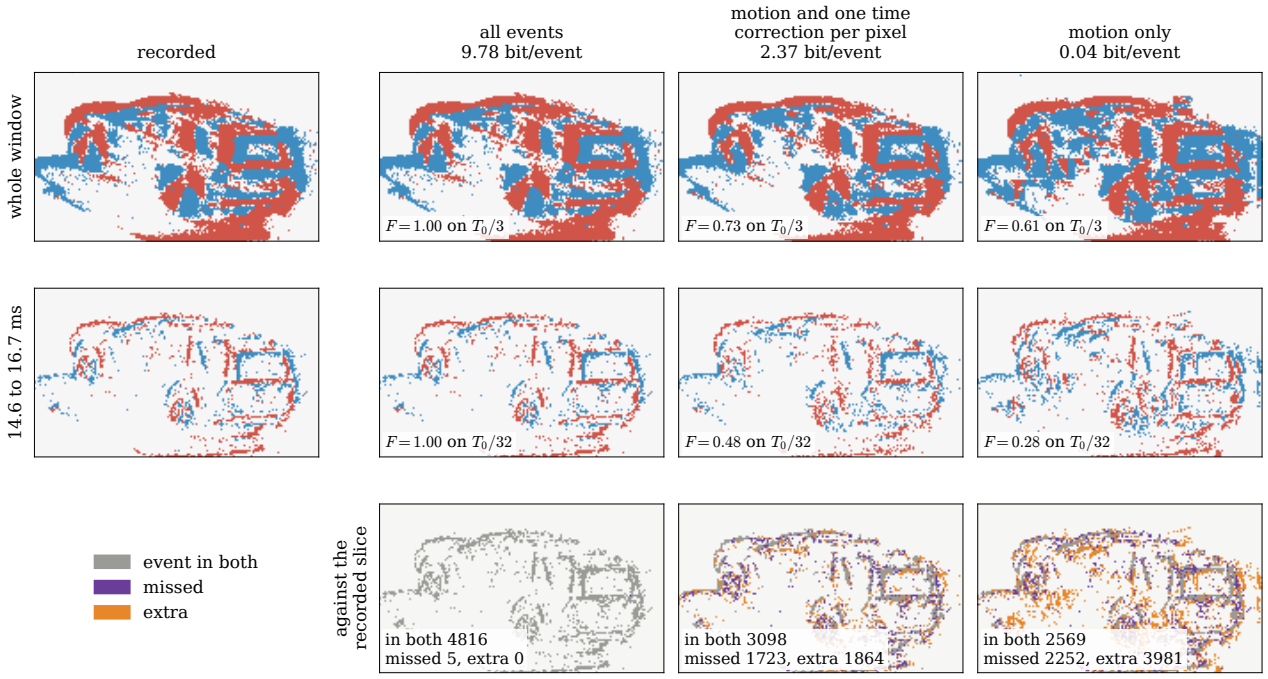


Fig. 5. A moving car in eTraM by day, as recorded and as received from three codes. Top: the whole window; middle: a slice of $T_0/16$; signed event counts, red for positive polarity. Bottom: the slice against the recorded one.

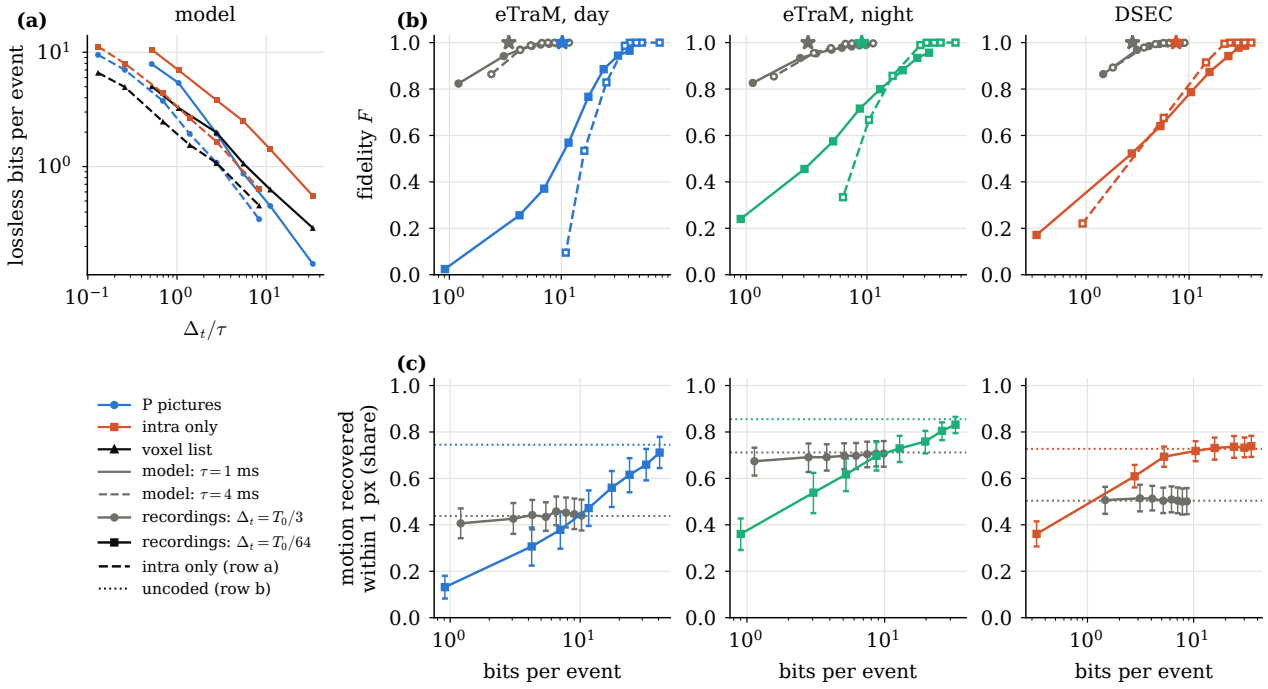


Fig. 6. x265 as a case. (a) Model, lossless bits per event, $n = 3$. (b) Recordings, fidelity against bits per event on the grids of $T_0/3$ (grey) and $T_0/64$; star: voxel list. (c) Share of squares whose motion the P-picture decode recovers within one pixel per window, decodes without events counted as failures; dotted: uncoded counts on the same grid.

The quantizer then removes events instead of coarsening them. At $T_0/64$ and QP 20 or below, 91% to 99% of the errors of a decode are voxels of one event that it empties, and 1% to 7% are events that it adds. The P pictures lose more of them than intra pictures at the same QP, so at equal fidelity the intra pictures are as cheap or cheaper: to reach $F = 0.9$ at $T_0/64$ they need 1.00, 0.89 and 0.72 times the bits of the P pictures by day, at night and on DSEC (Fig. 6b). On the grid of $T_0/3$, where voxels hold several events, the quantizer coarsens counts instead, and F stays above 0.82 down to about one bit per event. The two grids answer different timing requirements: F on $T_0/3$ ignores timing differences within 11 ms, so its curves

give the cost of a looser requirement and not that of a better coder.

The motion task. The fine grid serves the task best when it is coded finely. Uncoded, the counts on $T_0/64$ let 75%, 86% and 73% of the squares be recovered, against 44%, 71% and 50% on $T_0/3$ (Fig. 6c). As the rate falls, the decode on $T_0/64$ empties squares: by day the recovered share falls from 0.71 at 41 bits per event to 0.13 at 0.9, with 49% of the squares left without events; at night it falls from 0.83 to 0.36 and on DSEC from 0.74 to 0.36. On $T_0/3$ the share stays at its uncoded value at every QP. Below a crossing that lies between 1.5 and 10 bits per event depending on the stratum, the coarse grid therefore serves the task better. Counting a decode without events as a failure matters here: at QP 24 by day, 29% of the squares are left without events, and the median error over all squares is 3.4 pixels per window, against 1.4 when only the squares with events enter.

VII. Conclusion

The pixels on the path of a moving edge share the edge and its motion, and each holds a threshold phase and a jitter of its own. Given the edge and the motion, a timing tolerance decides which of these private quantities a code must describe: below the jitter variance every event time, up to $D_A = \sigma^2 + \tau^2/12$ one number per pixel, and beyond it nothing, so that a description of the motion of $O(\ln M)$ nats replaces a description of order M. The event spacing sets the scale of the tolerance needed for regeneration and of the time bins over which motion-aligned events remain predictive. A sender that moves events along the motion keeps the phase of the source pixel, and a controlled model shows that the phase alone then makes moved counts miss on bins shorter than the event spacing. Recordings show no detectable phase lock on the regular trains where the test applies, and an HEVC coder on count pictures shows the consequences on fine grids: prediction helps against intra pictures but not against a list of the counts, and lossy decodes remove single events. The theory points to a sender that estimates the edge from the events and sends it with the motion. Its scope is a ramp edge in uniform passage, and texture, noise events and threshold mismatch remain to be modeled.

Appendix A
Remarks on the Model

Remark 3 (Accuracy of the uniform passage). For an affine flow (4), $\dot\nu_t = -\Lambda^\top \nu_t$ and $\ddot z_i = \langle \nu_t, \Lambda v(i)\rangle$, hence $|\ddot z_i| \le \kappa_i |\dot z_i|$ with $\kappa_i = \|\Lambda\|/|\cos\vartheta_i|$, where ϑ_i is the angle between the velocity $v(i)$ and the normal ν_t. Take r_i as the slope of z_i at t_i. Over a passage of duration $P_i = n\tau_i$, the event times then deviate from (10) by at most $\kappa_i P_i^2/2$, to first order in $\kappa_i P_i$. For a generator of norm $0.3\,\mathrm{s}^{-1}$, a passage of 5 ms and motion along the normal this is $4\,\mu$s, three orders of magnitude below the passage time. The bound grows without limit for edges that are nearly parallel to their motion. Such edges pass few pixels per unit time and contribute few events.

Remark 4 (Event count under a change of scale). An edge segment emits n events for every pixel that it passes, so its event rate is n times the area that it sweeps per unit time, that is n times its length times its mean normal speed. Under a similarity flow about a fixed point, growth by a factor ζ makes the segment ζ times longer and ζ times faster, and the rate grows as ζ^2. For an object that approaches at constant speed, the image speed itself grows as ζ^2 and the rate as ζ^3. A regeneration that moves a fixed set of events keeps their number. Under a change of scale it must also change the number of events, or regenerate them from the edge.

Appendix B
Proofs of Lemmas 1 and 2

Proof of Lemma 1. The reference starts in $a_i + C\mathbb{Z}$ by the definition of the phase. By (A1) it changes by $\pm C$, so it stays in this set, and an event requires $\ell_i = \mu_i \pm C$. □

Proof of Lemma 2. By (A2) and (A2′), ℓ_i is nondecreasing, equal to 0 before t_i, to $\gamma r_i(t - t_i)$ during the passage, and to V after it. Since $\mu_i \le 0 = \ell_i$ before the edge arrives and $\mu_i = a_i - C$, the events occur when ℓ_i reaches the levels $a_i + (k-1)C$, $k = 1, 2, \dots$, as long as the level is at most $V = nC$. Because $0 < a_i \le C$, exactly the levels $k = 1, \dots, n$ are reached. Solving $\gamma r_i(t - t_i) = a_i + (k-1)C$ gives (10), and a_i uniform on $(0, C]$ makes A_i uniform on $(0, \tau_i]$. □

Appendix C
Proof of Theorem 1

By Lemma 2, $T_i = c_i + A_i \mathbf{1} + \varepsilon_i$ with c_i a function of the template and the flow, $\mathbf{1}$ the all-ones vector, and $\varepsilon_i \sim \mathcal{N}(0, \sigma^2 I_n)$. Independence over i follows from (A3) and (A4). Let Q be an orthogonal matrix whose first row is $\mathbf{1}^\top/\sqrt{n}$. Then $Q(T_i - c_i) = (X_i, W_i)$ with

$$X_i = \sqrt{n}\, A_i + \varepsilon_i^{\parallel}, \qquad W_i \sim \mathcal{N}(0, \sigma^2 I_{n-1}), \tag{34}$$

where $\varepsilon_i^{\parallel} \sim \mathcal{N}(0, \sigma^2)$. The variables X_i and W_i are independent, because $Q\varepsilon_i$ has independent coordinates and A_i is independent of ε_i. Differential entropy is invariant under translations and orthogonal maps, so $h(T_i) = h(X_i) + h(W_i)$, which is (11).

Let f be the density of T_i and f_δ its average over the cells of the clock lattice. A cell has probability $\delta^n f_\delta$, so $H(Y_i \mid g, \phi) = n\ln(1/\delta) - \int f \ln f_\delta$ and $e_\delta = \int f \ln(f/f_\delta)$ is a relative entropy, which is nonnegative. The density f is a mixture over A_i of $\mathcal{N}(m, \sigma^2 I_n)$ densities and f_δ is the same mixture of their cell averages. By the joint convexity of relative entropy, e_δ is at most its largest value for a single $\mathcal{N}(m, \sigma^2 I_n)$ density, which is n times the value for a scalar normal density and depends only on δ/σ and on the offset of m in a cell. It tends to zero with δ/σ, uniformly in the offset [26, Ch. 8].■

Appendix D
Proof of Theorem 2

The distortion (14) is invariant under the translation by c_i and the orthogonal map Q, so R_i is the rate-distortion function of the independent components (X_i, W_i) of (34) under the total squared error nD. Their variances are s_i^2 and σ^2, the latter

$n-1$ times. We drop the index i and write η for $\eta(\sqrt{n}\,\tau_i,\sigma)$ and $\ln^+ x = [\ln x]^+$.

Lower bound (17) *and* (18). Let $(\hat X,\hat W)$ have component errors d_X and $d_1,\dots,d_{n-1}$ with $d_X+\sum_j d_j \le nD$. Independence of the components gives $I(X,W;\hat X,\hat W)\ge I(X;\hat X)+\sum_j I(W_j;\hat W_j)$. For a Gaussian component, $I(W_j;\hat W_j)\ge \frac12\ln^+(\sigma^2/d_j)$, and $I(X;\hat X)\ge h(X)-h(X-\hat X)\ge \eta-\frac12\ln(2\pi e\,d_X)$. The function $\ln^+(\sigma^2/d)$ is convex in d, so the Gaussian terms are smallest when all d_j equal a common d_W, and $d_W>\sigma^2$ brings no benefit. For $n\ge 2$ this gives

$$R(D)\;\ge\;\min_{d_W}\Big[\eta-\tfrac12\ln\big(2\pi e\,(nD-(n-1)d_W)\big) + \tfrac{n-1}{2}\ln\frac{\sigma^2}{d_W}\Big], \tag{35}$$

with the minimum over $0<d_W\le\sigma^2$, $d_W<nD/(n-1)$. The bracket is convex in d_W and its derivative vanishes only at $d_W=D$. The minimum is at $d_W=D$ when $D\le\sigma^2$, where it equals the right side of (17), and at $d_W=\sigma^2$ otherwise, where it equals $U(D)-\Gamma$. For $n=1$ there is no W and the same values follow with $d_X=D$.

Equality for $D\le\sigma^2$. Write $\varepsilon=\varepsilon'+Z$ with independent $\varepsilon'\sim\mathcal N(0,(\sigma^2-D)I_n)$ and $Z\sim\mathcal N(0,DI_n)$, and let $\hat T = c+A\mathbf 1+\varepsilon'$. Then $T=\hat T+Z$ with Z independent of $\hat T$, the distortion is D, and $I(T;\hat T)=h(T)-h(Z)$, which is the lower bound.

Upper bound for $\sigma^2<D<D_A$. Spend no rate on W, at error σ^2 per component, and describe X at error $d=nD-(n-1)\sigma^2\in(\sigma^2,s^2)$ through $\hat X=(1-d/s^2)(X-\mathbb EX+Z')+\mathbb EX$, with Z' an independent normal variable of variance $s^2d/(s^2-d)$. The error is d, and $I(X;\hat X)=I(X;X+Z')\le\frac12\ln(s^2/d)$, because a normal input maximizes the mutual information across an additive normal noise channel at a given input variance [26].

Lower bound (21) *and zero rate.* For a variable Ξ with $\ln\mathbb E\,e^{\lambda(\Xi-\mathbb E\Xi)}\le\lambda^2\varsigma^2/2$ for all λ under a law P, and any law Q, the nonnegativity of the relative entropy between Q and the law with density proportional to $e^{\lambda\Xi}$ with respect to P gives $D(Q\|P)\ge\lambda(\mathbb E_Q\Xi-\mathbb E_P\Xi)-\lambda^2\varsigma^2/2$, and the best λ gives $(\mathbb E_Q\Xi-\mathbb E_P\Xi)^2\le 2\varsigma^2D(Q\|P)$. With P the law of X and Q its conditional law given $\hat T$, averaging over $\hat T$ yields $\operatorname{Var}\mathbb E[X\mid\hat T]\le 2\varsigma^2 I(X;\hat T)$. The uniform part of X has width $a=\sqrt{n}\,\tau$ and satisfies the condition with $a^2/12$, since $\ln(\sinh x/x)\le x^2/6$, so X satisfies it with $\varsigma^2=s^2$, and each W_j with $\sigma^2\le s^2$. Since the conditional mean minimizes the squared error and the components are independent,

$$\begin{aligned} nD &\ge \mathbb E\|T-\mathbb E[T\mid\hat T]\|^2\\ &= nD_A-\operatorname{Var}\mathbb E[X\mid\hat T]-\sum_j\operatorname{Var}\mathbb E[W_j\mid\hat T]\\ &\ge nD_A-2s^2I(T;\hat T), \end{aligned} \tag{36}$$

which is (21). With no description the best output is the mean, which is (22), at distortion $\frac1n(s^2+(n-1)\sigma^2)=D_A$, so (19) holds. ∎

Appendix E
Proof of Theorem 3

The vectors T_i are independent and the distortion is additive over i. For any $\hat T$ with average distortion at most D, let $D_i=\mathbb E\,d(T_i,\hat T_i)$. Then $I\big((T_i)_i;(\hat T_i)_i\big)\ge\sum_i I(T_i;\hat T_i)\ge\sum_i R_i(D_i)$ with $\frac1M\sum_i D_i\le D$. Conversely, independent test channels whose rates are within any given margin of $R_i(D_i)$ come within M times that margin of the sum. Each R_i is convex, nonincreasing and continuous, and tends to infinity as $D_i\to0$ by (17), so the minimum in (23) exists.

For $D<\sigma^2$, every R_i is differentiable at D with the derivative $-n/(2D)$ by (17), and it is convex on $(0,\infty)$. Hence $R_i(D_i)\ge R_i(D)-\frac{n}{2D}(D_i-D)$ for every $D_i>0$, and summing over i with $\sum_i(D_i-D)\le0$ shows that the allocation $D_i=D$ is optimal. This gives (24) for $D<\sigma^2$, and the case $D=\sigma^2$ follows by continuity.

By Theorem 2, $\sum_i R_i(D_i)=0$ if and only if $D_i\ge D_{A,i}$ for every i, and such an allocation with average at most D exists if and only if $D\ge\frac1M\sum_i D_{A,i}=\bar D_A$. Finally, (21) and $s_i^2\le n\tau_{\max}^2/12+\sigma^2$ give $\sum_i R_i(D_i)\ge n\sum_i[D_{A,i}-D_i]^+/(2(n\tau_{\max}^2/12+\sigma^2))$, and $\sum_i[D_{A,i}-D_i]^+\ge M(\bar D_A-D)$, which is (26). ∎

Appendix F
Proof of Proposition 1

Divide side j of the box into the smallest number of equal cells of length at most $\varsigma_j=\sqrt{\xi}\,\sigma/(q\,S_jM^{\beta_j})$, and let $\hat\theta$ be the center of the cell that contains θ. Side j has at most $L_jM^{\alpha_j}/\varsigma_j+1$ cells, so the index of $\hat\theta$ takes $\sum_j\ln\big(L_jS_jq\,M^{\alpha_j+\beta_j}/(\sqrt{\xi}\,\sigma)+1\big)$ nats, which is (27). The decoder outputs $\hat T_{i,k}=\bar t_{i,k}(\hat\theta)$. The segment from θ to $\hat\theta$ lies in the box, so by the mean value theorem $e_{i,k}=\bar t_{i,k}(\theta)-\bar t_{i,k}(\hat\theta)$ satisfies $|e_{i,k}|\le\sum_j S_jM^{\beta_j}\varsigma_j/2\le\sqrt{\xi}\,\sigma/2$. By (10), $T_{i,k}-\bar t_{i,k}(\theta)=A_i-\tau_i/2+\varepsilon_{i,k}$ has mean zero given θ and variance $D_{A,i}$, and $e_{i,k}$ is a function of θ. Hence $\mathbb E\,(T_{i,k}-\hat T_{i,k})^2=D_{A,i}+\mathbb E\,e_{i,k}^2\le D_{A,i}+\xi\sigma^2/4$, and the average over the path is at most $\bar D_A+\xi\sigma^2/4\le(1+\xi)\bar D_A$. ∎

Appendix G
Proof of Proposition 2

By (10) and (A4), $T_{i,k}-\tilde T_{i,k}=(A_i-A_j)+(k-1)(\tau_i-\tau_j)+\varepsilon_{i,k}-\varepsilon_{j,k}$. The phases and the jitter of the two pixels are independent, so this difference has the mean $(k-\frac12)(\tau_i-\tau_j)$ and the variance $\tau_i^2/12+\tau_j^2/12+2\sigma^2=D_{A,i}+D_{A,j}$. Its mean square is the variance plus the squared mean, and $\frac1n\sum_{k=1}^n(k-\frac12)^2=(4n^2-1)/12$. ∎

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